Sharp first-order energy correction under an Eulerian tidal bound for finite-wall Alcubierre profiles

Tatsuhiko Takimoto

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Abstract

We consider the stationary Alcubierre metric with speed $v = 1/10$ and globally C^2, monotone radial profiles that match constant interior and exterior values at radii 1 and $R \in [3/2, 3]$. For the Eulerian normal, we bound the spectral norm of the electric part of the full Riemann tensor by ηv^2 at every point of the wall. If $J_R^*(\eta)$ denotes the infimum of $\int_1^R r^2 f'(r)^2 \, \mathrm{d}r$ over this class, then

$$\eta \left[J_R^*(\eta) - \frac{R}{R-1} \right] \longrightarrow \frac{1}{3R(R-1)^3}$$

uniformly in R as $\eta \to \infty$. An axial necessary condition gives a lower bound for every admissible profile. A normalized, smoothed ramp construction gives an upper bound satisfying the tidal constraint at all angles. The errors in the scaled lower and upper bounds are respectively $O(\eta^{-1/2})$ and $O(\eta^{-1})$, uniformly in R. No minimizing profile or uniform higher-derivative bound is assumed. Eighteen exact rational enclosures illustrate the bounds.

1 Introduction

Alcubierre's metric provides a concrete setting in which a prescribed spacetime geometry determines an Eulerian energy density [1]. Pfenning and Ford related curvature scales, wall thickness and quantum-inequality restrictions [2]. Lobo and Visser used a volume integral to quantify negative energy and distinguished that quantity from ADM mass [3]. The present problem concerns the cost of imposing a pointwise tidal bound on a fixed finite wall.

Shape optimization is already part of this literature. Bobrick and Martire [4, arXiv v2, App. A.3] vary the radial energy functional and obtain a harmonic profile. Natário's divergence-free shift construction [5] leads to a different admissible class. For that class, Bolívar, Vasilev and Abellán report a sharp Eulerian energy minimum and a thin-wall asymptotic through a Stokes-flow formulation [6]. Here the wall width stays in $[1/2, 2]$. The small scale instead arises from endpoint layers as the allowed electric tidal norm increases.

General shift constructions by Lentz [7] and Fell and Heisenberg [8], and the curvature and observer analysis of Santiago, Schuster and Visser [9], provide further context. In particular, Eulerian density and energy conditions involving other observers are different questions. Warp Factory [10] provides computational tools for geometry and energy-condition analysis; Le reports observer-robust verification methods in Warpax [11]. Our conclusion is an analytic asymptotic for a specified scalar profile class. Higher-order variational expansions have a general framework [12]; the proof here uses direct inequalities and an explicit recovery family.

The quantity determined below is the first correction to the harmonic energy floor under an all-angle electric-Riemann norm constraint. The outer endpoint forces a linear loss in the rescaled profile. At the inner endpoint, the vanishing shift permits a much thinner transition. This asymmetry determines the coefficient.

2 The variational problem and the main result

Work in units $G = c_{\text{light}} = 1$, fix $v = 1/10$, and write

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \big(\mathrm{d}z + v(1 - f(r))\,\mathrm{d}t\big)^2, \qquad r = (x^2 + y^2 + z^2)^{1/2}. \tag{2.1}$$

For $R \in [3/2, 3]$, the profile f is globally C^2 and satisfies

$$f = 1 \text{ on } [0, 1], \qquad f = 0 \text{ on } [R, \infty), \qquad f' \le 0.$$

Thus $f' = f'' = 0$ at both matching radii. There is no uniform bound on derivatives of order three or higher. Set

$$h = 1 - f, \qquad p = h' = -f', \qquad n = \partial_t - vh\,\partial_z.$$

The spatial vectors $\partial_x, \partial_y, \partial_z$ form an orthonormal frame perpendicular to n. Let

$$E_{ij} = R_{i\alpha j\beta} n^\alpha n^\beta, \qquad \left\| E/v^2 \right\|_2 \le \eta \quad \text{at every wall radius and angle.} \tag{2.2}$$

Here E is the electric part of the full Riemann tensor, and $\|\cdot\|_2$ is the Euclidean operator norm. An overall reversal of the curvature sign does not affect the constraint. The unit lapse makes the Eulerian normal geodesic, so E is its geodesic-deviation tensor. For any fixed nonzero v, both E/v^2 and J below are independent of v; the choice $v = 1/10$ fixes the energy conversion in (3.5).

Let $\mathcal{A}_R(\eta)$ be the class just defined, for $\eta \ge 128$, and put

$$J(f) = \int_1^R r^2 p(r)^2\,\mathrm{d}r, \qquad J_R^*(\eta) = \inf_{f \in \mathcal{A}_R(\eta)} J(f). \tag{2.3}$$

For later use, define

$$C = \frac{R}{R-1}, \qquad a = \frac{1}{R(R-1)}, \qquad K_R = \frac{R^2 a^3}{3} = \frac{1}{3R(R-1)^3}. \tag{2.4}$$

In this range $3/2 \le C \le 3$, $1/6 \le a \le 4/3$, and $K_R \le 16/9$.

Theorem 1. *For every $R \in [3/2, 3]$ and $\eta \ge 128$, the class $\mathcal{A}_R(\eta)$ is nonempty and*

$$\eta(J_R^*(\eta) - C) \ge L_R(\eta) := \frac{(R - a/\eta)^2 a^3}{3}\left(1 - 2\sqrt{a/\eta}\right). \tag{2.5}$$

The constants $B_- = 5$, $B_+ = 61000$, and $\eta_0 = 128$, independent of R, satisfy, for all $\eta \ge \eta_0$,

$$K_R - B_- \eta^{-1/2} \le \eta(J_R^*(\eta) - C) \le K_R + B_+ \eta^{-1}. \tag{2.6}$$

Consequently,

$$\lim_{\eta \to \infty} \sup_{R \in [3/2,3]} \left| \eta\left[J_R^*(\eta) - \frac{R}{R-1}\right] - \frac{1}{3R(R-1)^3} \right| = 0. \tag{2.7}$$

No attainment of the infimum is required.

The coefficient is sharp because a lower bound for the entire admissible class and an admissible family have the same first-order limit. The rates in (2.6) are estimates for the bounds; we do not assert that they are optimal convergence rates.

3 Curvature and the energy functional

Use the convention

$$R^a{}_{cmn} = \partial_m \Gamma^a_{nc} - \partial_n \Gamma^a_{mc} + \Gamma^a_{mk}\Gamma^k_{nc} - \Gamma^a_{nk}\Gamma^k_{mc}.$$

For a stationary shift $b\,\partial_z$, set $d = \nabla b$, $H = \nabla^2 b$ and $e = (0,0,1)^T$. The nontrivial inverse metric components are $g^{tt} = -1$, $g^{tz} = b$, and $g^{zz} = 1 - b^2$. Direct evaluation of the Christoffel symbols and contraction with $n = (1,0,0,-b)$ give

$$\begin{aligned} E = &-\frac{b}{2}\big((He)e^T + e(He)^T\big) - \frac{3}{4}dd^T \\ &- \frac{b_z}{4}(de^T + ed^T) + \frac{|d|^2}{4}ee^T. \end{aligned} \tag{3.1}$$

This is a specialization of the flat-slice curvature identities discussed in [9, arXiv v2, Sec. 3.1]. For example, it also follows by taking $\mathsf{K} = (de^T + ed^T)/2$ and expanding

$$E = -\mathcal{L}_{be}\mathsf{K} + \mathsf{K}^2, \qquad \mathcal{L}_{be}\mathsf{K} = b\,\partial_z\mathsf{K} + d(e^T\mathsf{K}) + (\mathsf{K}e)d^T$$

in the present stationary convention.

Substitute $b = vh$. Rotational symmetry about the z-axis allows the radial unit vector to be written as $(s,0,u)$, with $u \in [-1,1]$ and $s = \sqrt{1-u^2}$. Then

$$E/v^2 = \begin{pmatrix} A & 0 & B \\ 0 & 0 & 0 \\ B & 0 & D \end{pmatrix}, \quad \begin{aligned} A &= -\tfrac{3}{4}p^2(1-u^2), \\ B &= -\tfrac{h}{2}(p' - p/r)su - p^2 su, \\ D &= -hp'u^2 - \tfrac{hp}{r}(1-u^2) + \tfrac{1-5u^2}{4}p^2. \end{aligned} \tag{3.2}$$

On the axis, $E_{zz}/v^2 = -hp' - p^2 = -(hp)'$. Thus every admissible profile satisfies

$$|(hp)'| \le \eta. \tag{3.3}$$

For the upper construction we need an estimate valid at every angle:

$$\big\|E/v^2\big\|_2 \le |hp'| + \frac{hp}{r} + \frac{3}{2}p^2. \tag{3.4}$$

Here $h, p \ge 0$. The coefficient of hp' has norm $(u^2 + |u|)/2 \le 1$. The coefficient of hp/r has norm $(\sqrt{1-u^2} + 1 - u^2)/2 \le 1$. The remaining terms in (3.2) have total norm at most $p^2[3/4 + |u|(1+|u|)/4 + 1/4] \le 3p^2/2$. The axial condition is used only as a necessary condition; it is not substituted for the all-angle constraint.

The Einstein contraction, with $G_{ab} = 8\pi T_{ab}$, is

$$G(n,n) = -\frac{b_x^2 + b_y^2}{4}, \qquad \rho = T(n,n) = -\frac{v^2 p^2 \sin^2\theta}{32\pi}.$$

The Hamiltonian expression $\big((\operatorname{tr}\mathsf{K})^2 - \operatorname{tr}(\mathsf{K}^2)\big)/2$ gives the same contraction. Since the spatial slice is Euclidean,

$$\mathcal{E} := \int_{\Sigma_t} \rho \,\mathrm{d}V = -\frac{v^2}{12}\int_1^R r^2 p^2 \,\mathrm{d}r = -\frac{J}{1200}. \tag{3.5}$$

Thus J is proportional to the magnitude of this negative Eulerian volume integral. Its interpretation as a slice integral follows [1, 3]; it is not an ADM mass.

4 A lower bound for every admissible profile

Let $p_0(r) = C/r^2$. We have $\int_1^R p_0\,\mathrm{d}r = 1$ and $\int_1^R r^2 p_0^2\,\mathrm{d}r = C$. Since every profile also has $\int_1^R p\,\mathrm{d}r = 1$, expansion of the square gives the exact identity

$$J - C = \int_1^R r^2 (p - p_0)^2\,\mathrm{d}r \geq 0. \tag{4.1}$$

The reference function p_0 does not satisfy the endpoint conditions.

Put $w = hp$. Equation (3.3) and $w(R) = 0$ imply

$$hp(r) \leq \eta(R - r). \tag{4.2}$$

First consider profiles for which $J \leq C + 4/\eta < 4$. Since $a/\eta \leq 1/96 < 1/2 \leq R - 1$, the interval $[R - a/\eta, R]$ lies inside the wall. On it, $\int_r^R t^{-2}\,\mathrm{d}t = (R - r)/(rR) \leq R - r$, so Cauchy–Schwarz gives

$$1 - h(r) = \int_r^R p(t)\,\mathrm{d}t \leq \left(J \int_r^R t^{-2}\,\mathrm{d}t \right)^{1/2} \leq 2\sqrt{R - r} \leq 2\sqrt{a/\eta} =: \beta < 1.$$

In fact $\beta \leq 2\sqrt{(4/3)/128} < 0.21$. On the smaller interval $0 \leq t = R - r \leq a(1 - \beta)/\eta$, the ordering

$$p(r) \leq \frac{\eta t}{1 - \beta} \leq a \leq p_0(r)$$

permits squaring $p_0 - p \geq a - \eta t/(1 - \beta) \geq 0$. Restricting (4.1) to this interval therefore yields

$$\begin{aligned} \eta(J - C) &\geq \eta(R - a/\eta)^2 \int_0^{a(1-\beta)/\eta} \left(a - \frac{\eta t}{1 - \beta} \right)^2 \mathrm{d}t \\ &= \frac{(R - a/\eta)^2 a^3 (1 - \beta)}{3} = L_R(\eta). \end{aligned}$$

The integration interval lies inside the wall. If $J > C + 4/\eta$, then $\eta(J - C) > 4 > K_R \geq L_R(\eta)$, so the same conclusion holds. Taking the infimum proves (2.5). The bounds on R and a imply

$$L_R(\eta) = K_R + O(\eta^{-1/2}) \tag{4.3}$$

uniformly in R. More explicitly,

$$0 \leq K_R - L_R(\eta) \leq K_R \left(\frac{2a}{R\eta} + 2\sqrt{a/\eta} \right) \leq 5\eta^{-1/2}.$$

For the last step use $K_R \leq 16/9$, $2a/R \leq 16/9$, $1/\sqrt{128} < 1/11$, and $4/\sqrt{3} < 7/3$. This argument applies to all admissible C^2 profiles, including those far from the energy floor.

5 An admissible recovery family

5.1 Construction and normalization

For $\eta \geq 128$, set

$$\gamma = \eta - 64 \geq 64, \qquad \delta = \eta^{-3}, \qquad \epsilon = \eta^{-4} < \delta.$$

Define on $[1, R]$

$$q_\eta(r) = \min \left\{ \frac{C}{r^2}, \frac{C}{\delta}(r - 1 - \delta)_+, \gamma(R - \delta - r)_+ \right\} \tag{5.1}$$

and extend it by zero outside the wall. The three active pieces are an increasing line, C/r^2, and a decreasing line. Their junctions satisfy

$$\begin{aligned} (\alpha - 1 - \delta)\alpha^2 &= \delta, \qquad 1 + \delta \le \alpha \le 1 + 2\delta, \\ (R - \delta - \zeta)\zeta^2 &= C/\gamma, \qquad R - \delta - C/\gamma \le \zeta \le R - \delta. \end{aligned} \tag{5.2}$$

The two intervals are disjoint. The inner line increases while C/r^2 decreases. The function $(R - \delta - r)r^2$ is unimodal, is greater than C/γ at $r = 1$, and is decreasing on the outer junction interval, because

$$\frac{C}{\gamma} \le \frac{3}{64} < \frac{R - \delta}{3}, \qquad R - 1 - \delta \ge \frac{1}{2} - 128^{-3} > \frac{3}{64}.$$

Also $(r - 1 - \delta)r^2$ is strictly increasing on the inner interval. These inequalities give the unique stated junctions and decomposition. In particular,

$$0 \le q_\eta \le C, \qquad \mathrm{TV}(q_\eta) \le 2C, \qquad -\gamma \le q_\eta' \le C/\delta \quad \text{a.e.}$$

Here the derivative on the harmonic piece is at least $-6 \ge -\gamma$.

Write $I = \int_1^R q_\eta \, \mathrm{d}r$. The missing mass at the inner end is at most $2C\delta$. At the outer end, the zero strip loses at most $C\delta$; comparison with a ramp under the constant C bounds the remaining loss by $C^2/(2\gamma)$. Hence

$$1 - I \le 3C\delta + \frac{C^2}{2\gamma} \le \frac{8}{\gamma}, \qquad \frac{7}{8} \le I \le 1. \tag{5.3}$$

The last estimate follows from $C \le 3$ and $9\eta^{-3} \le 7/[2(\eta - 64)]$.

Use the nonnegative even kernel

$$\kappa(t) = \frac{15}{16}(1 - t^2)^2 \mathbf{1}_{\{|t| \le 1\}}, \qquad \kappa_\epsilon(t) = \epsilon^{-1}\kappa(t/\epsilon).$$

It has mass one and first absolute moment $5/16$. Define

$$p_\eta = \frac{q_\eta * \kappa_\epsilon}{I}, \qquad h_\eta(r) = \int_1^r p_\eta(t) \, \mathrm{d}t, \qquad f_\eta = 1 - h_\eta. \tag{5.4}$$

Extend h_η by 0 and 1 on the inner and outer constant regions. Since $\epsilon < \delta$, the support of p_η stays strictly inside the wall. Since q_η is Lipschitz, $(q_\eta * \kappa_\epsilon)' = q_\eta' * \kappa_\epsilon$ is continuous; nonnegativity and unit mass of the kernel preserve the a.e. derivative bounds. Hence f_η is globally C^2, is monotone, and has both required zero derivatives at the endpoints. Its normalization is exact.

5.2 The tidal bound at all wall points

Put $A = C/I \le 24/7 < 4$. Then

$$p_\eta \le A, \qquad -\gamma/I \le p_\eta' \le A/\delta.$$

The derivative q_η' is positive only on $(1 + \delta, \alpha)$ and is nonpositive a.e. on $[\alpha, R]$. If $p_\eta'(r) > 0$, the convolution window meets $(1 + \delta, \alpha)$, so $r \le 1 + 2\delta + \epsilon$. There

$$h_\eta \le A(2\delta + \epsilon) \le 3A\delta, \qquad h_\eta p_\eta' \le 3A^2 < 48.$$

If $p_\eta' < 0$, then

$$|h_\eta p_\eta'| \le \frac{\gamma}{I} \le \frac{\gamma}{1 - 8/\gamma} \le \gamma + \frac{64}{7} < \eta - 54.$$

The remaining terms in (3.4) satisfy

$$h_\eta p_\eta / r + \tfrac{3}{2} p_\eta^2 < 4 + 24 = 28.$$

Thus the right side of (3.4) is less than $76 \le \eta$ on the increasing part and less than $\eta - 26$ on the decreasing part; the case $p_\eta' = 0$ is immediate. This proves $f_\eta \in \mathcal{A}_R(\eta)$ for every parameter pair in the stated range, without angular sampling.

5.3 The first correction, uniformly in R

Let $J_q = \int_1^R r^2 q_\eta^2 \,\mathrm{d}r$. The inner transition changes mass and energy by $O(\delta)$, uniformly in R. For the outer layer, set

$$x = \gamma(R - r), \qquad \lambda = \gamma\delta = O(\gamma^{-2}), \qquad x_0 = \gamma(R - \zeta) \in [\lambda, \lambda + C] \subset [0, 4].$$

At the junction,

$$x_0 - \lambda = \frac{C}{(R - x_0/\gamma)^2}.$$

The derivative of $p_0 = C/r^2$ is bounded in absolute value by 6 throughout the wall. Therefore, on this bounded x-interval,

$$p_0(R - x/\gamma) = a + O(\gamma^{-1}), \quad (R - x/\gamma)^2 = R^2 + O(\gamma^{-1}), \quad x_0 = a + O(\gamma^{-1}) \tag{5.5}$$

uniformly in R. For example, each of the first two errors is at most $24/\gamma$, and $|x_0 - a| \le \lambda + 24/\gamma$. On the outer piece $q_\eta = (x - \lambda)_+$. Changing the integration endpoint from x_0 to a and using (5.5) gives

$$1 - I = \frac{1}{\gamma}\int_0^a (a - x)\,\mathrm{d}x + O(\gamma^{-2}) = \frac{a^2}{2\gamma} + O(\gamma^{-2}), \tag{5.6}$$

$$C - J_q = \frac{R^2}{\gamma}\int_0^a (a^2 - x^2)\,\mathrm{d}x + O(\gamma^{-2}) = \frac{2R^2a^3}{3\gamma} + O(\gamma^{-2}). \tag{5.7}$$

For explicit remainder bounds, there are four contributions. The inner losses are at most 6δ in mass and 18δ in energy. In the outer integral, variation of p_0 contributes at most $96/\gamma$ in mass; variation of r^2 and p_0^2 contributes $(1536 + 3744)/\gamma$ in energy. The ramp shift obeys

$$|(x - \lambda)_+ - x| \le \lambda, \qquad |(x - \lambda)_+^2 - x^2| \le 2x\lambda;$$

its contributions are at most 4λ and 288λ. Finally, moving the endpoint to a contributes at most $4(\lambda + 24/\gamma)$ and $144(\lambda + 24/\gamma)$. The outer contributions carry the factor $1/\gamma$. Since $\delta \le \gamma^{-3}$, $\lambda \le \gamma^{-2}$, and $\gamma \ge 64$, these estimates give

$$\left| I - 1 + \frac{a^2}{2\gamma} \right| \le \frac{200}{\gamma^2}, \qquad \left| J_q - C + \frac{2K_R}{\gamma} \right| \le \frac{9000}{\gamma^2}. \tag{5.8}$$

Here $|a - x| \le 4$, $|R^2(a^2 - x^2)| \le 144$, $|p_0^2 - (x - \lambda)_+^2| \le 16$, and $|p_0^2 - a^2| \le (3 + 4/3)24/\gamma = 104/\gamma$ justify the constants above on $0 \le x \le \max(a, x_0)$ where needed.

Using $I \ge 7/8$ and $C = R^2 a$, normalization now gives

$$\frac{J_q}{I^2} = C + \frac{Ca^2 - 2R^2a^3/3}{\gamma} + O(\gamma^{-2}) = C + \frac{K_R}{\gamma} + O(\gamma^{-2}). \tag{5.9}$$

For a numerical remainder constant, put $t = 1 - I \le 8/\gamma \le 1/8$. Then

$$0 \le (1 - t)^{-2} - 1 - 2t = \frac{t^2(3 - 2t)}{(1 - t)^2} \le 4t^2, \qquad \left| I^{-2} - 1 - \frac{a^2}{\gamma} \right| \le \frac{700}{\gamma^2}.$$

Expanding J_q/I^2 with (5.8) bounds the remainder in (5.9) by $14000/\gamma^2$: indeed

$$3 \cdot 700 + \frac{32}{9}\frac{16}{9} + \frac{32}{9}\frac{700}{64} + 9000\frac{64}{49} < 14000.$$

To control smoothing, the translation bound for total variation implies

$$\|q_\eta * \kappa_\epsilon - q_\eta\|_{L^1} \le \mathrm{TV}(q_\eta)\epsilon \int |t|\kappa(t)\,\mathrm{d}t \le \frac{5}{8}C\epsilon.$$

Both functions are bounded by C and supported inside $[1, R]$. Consequently

$$\left| \int_1^R r^2 (q_\eta * \kappa_\epsilon)^2 \, \mathrm{d}r - J_q \right| \le \frac{5}{4} R^2 C^2 \epsilon =: D_\eta. \tag{5.10}$$

Convolution preserves the mass. Since $D_\eta / I^2 \le (5/4)81(64/49)\gamma^{-4}$, we obtain the uniform bound

$$\left| J(f_\eta) - C - \frac{K_R}{\gamma} \right| \le \frac{15000}{\gamma^2}.$$

Finally, $\eta/\gamma \le 2$ gives

$$\eta(J(f_\eta) - C) \le K_R + \frac{64 K_R}{\gamma} + \frac{15000\eta}{\gamma^2} \le K_R + \frac{128 K_R + 60000}{\eta} \le K_R + \frac{61000}{\eta}.$$

Together with the explicit lower error estimate, this proves Theorem 1 with the stated constants.

6 Rational enclosures and reproduction

For rational R, η, all construction parameters are rational except the algebraic junctions. We enclose those junctions by 256 rational bisection steps in the monotone intervals (5.2). Set $a_0 = 1 + \delta$, $b_0 = R - \delta$, $S = \alpha - a_0$, and $T = b_0 - \zeta$. Exact integration gives

$$I = \frac{C}{2\delta} S^2 + C(1/\alpha - 1/\zeta) + \frac{\gamma}{2} T^2, \tag{6.1}$$

$$\begin{aligned} J_q = (C/\delta)^2 \left(\frac{a_0^2 S^3}{3} + \frac{a_0 S^4}{2} + \frac{S^5}{5} \right) + C^2 (1/\alpha - 1/\zeta) \\ + \gamma^2 \left(\frac{b_0^2 T^3}{3} - \frac{b_0 T^4}{2} + \frac{T^5}{5} \right). \end{aligned} \tag{6.2}$$

Each integral is monotone under enlargement of its interval because its integrand is nonnegative. Applying this fact separately to the three pieces produces rational bounds $I_- \le I \le I_+$ and $Q_- \le J_q \le Q_+$. Equation (5.10) then gives

$$\frac{Q_- - D_\eta}{I_+^2} \le J(f_\eta) \le \frac{Q_+ + D_\eta}{I_-^2}, \tag{6.3}$$

with a positive lower numerator in all tabulated cases. An integer-square-root construction gives a rational upper bound for $\sqrt{a/\eta}$, and hence a rational lower bound in (2.5).

Table 1 lists enclosures of $\Xi_R(\eta) = \eta[J_R^*(\eta) - C]$. Its upper endpoint comes from the explicit admissible family, not from locating an optimizer. All arithmetic preceding decimal presentation uses exact fractions; decimal endpoints are rounded outward. The exact rational endpoints and the generating script `reproduce.py` accompany this paper. An additional 120-digit, piecewise quadrature calculation reproduces the unsmoothed mass and energy inside their rational enclosures. The finite calculations check the formulas and construction; the uniform theorem follows from the preceding analytic bounds.

7 Scope and further questions

The result concerns the fixed one-component shift in (2.1), monotone radial C^2 profiles with exact endpoint matching, the stated Eulerian normal, and the electric part of the full Riemann tensor. The wall width does not tend to zero. The theorem neither bounds the remaining curvature components or other observers' tidal measurements nor addresses matter realizability, quantum inequalities, stability, or the satisfaction of energy conditions. Existence and uniqueness of a

Table 1: Outward-rounded rational enclosures of $\Xi_R(\eta) = \eta[J_R^*(\eta) - C]$. The lower endpoint is L_R from Theorem 1; the upper endpoint comes from the explicit admissible family in Section 5. The last column gives the proved limit.

R	η	Lower endpoint	Upper endpoint	K_R
3/2	128	1.395307386	3.658586574	16/9
3/2	512	1.590796077	2.039906892	
3/2	2048	1.685591794	1.836775343	
3/2	8192	1.732040918	1.792168268	
3/2	32768	1.755002098	1.781353645	
3/2	131072	1.766413588	1.778670395	
2	128	0.145264228	0.336030470	1/6
2	512	0.156097449	0.190686767	
2	2048	0.161418917	0.172082473	
2	8192	0.164052486	0.167988140	
2	32768	0.165362060	0.166995065	
2	131072	0.166014991	0.166748644	
3	128	0.012875360	0.028027553	1/72
3	512	0.013384811	0.015888803	
3	2048	0.013637562	0.014338348	
3	8192	0.013763409	0.013998455	
3	32768	0.013826195	0.013916111	
3	131072	0.013857553	0.013895684	

finite-η optimizer, sharper remainder estimates, and additional derivative constraints remain separate questions.

The proof is analytic. The accompanying Lean file records a weaker statement with existential error constants, rather than the explicit constants of Theorem 1. It formalizes neither the proof nor the metric-to-matrix derivation. Symbolic algebra and exact rational calculations support reproducibility. This draft has not undergone human peer review. The supplementary `VERIFICATION-RECORD.md` records the computational and derivational checks.

AI assistance. Claude and Codex were used for mathematical exploration, derivation, implementation, literature organization, and drafting. They are tools, not authors. The named author is responsible for the content and for decisions about publication.

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